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Modern AI, LLMs, and Agent Harnesses: A Field Guide for the Working Practitioner

Who this is for, and what it promises

If you have used a chatbot, written a few prompts, and now suspect there is a whole iceberg of jargon under the waterline, this guide is for you. The goal is not to turn you into a researcher. It is to get you fluent enough in the *current* conversation that you can use these tools well, follow the debates that working builders are actually having in 2026, and know where to dig when one topic grabs you.

So the bias here is practical. We will touch the theory where you need it to make sense of your own day-to-day experience, then move on. The questions driving this guide are the ones a user actually feels: Why does the model sometimes need me to paste in documents? Why are people suddenly running these things from a terminal? What is this "context window" everyone keeps blaming? And which techniques that I just learned are already on their way out?[92] By the end you should be able to do four concrete things: pick the right model for a job (inside one lab's lineup and across the whole field), prompt it well, understand why serious work needs tools and harnesses rather than a bare chatbot, and recognize the orchestration patterns where models start directing other models.

A note on the acronyms, because this field drowns in them: every one gets spelled out the first time it appears. **Artificial intelligence (AI)**, the broad term, mostly means one specific thing in this guide, the **large language model (LLM)**, a system trained on enormous amounts of text to predict and generate language.

The simplest map: models, interfaces, harnesses

Here is the whole field in one sentence. The last decade of AI is a story about three layers stacked on top of each other: **models** that grew more capable, **interfaces** that made those capabilities usable by normal people, and **harnesses** that let models act in the world through tools, memory, files, shells, browsers, and structured workflows.[1]

That layering is why AI felt like a lab curiosity in one era, a chatbot in the next, and a coding-and-agent platform after that. Nothing magical happened in a single year. Each layer unlocked the one above it.

A version that stays honest to the research but reads like plain English:

Classical deep learning made pattern recognition powerful. Transformers made language modeling general-purpose. Large language models made a single system usable for many tasks just by prompting it. Instruction tuning and preference tuning taught models to follow humans. Retrieval, tools, and structured outputs made them reliable enough for real applications. And agent harnesses gave them the ability to plan, act, observe, retry, and coordinate over time.[2]

The same progression explains how *your* experience of using AI changed:

Before about 2020, you trained or fine-tuned a model for one task. Around 2020 to 2022, you increasingly just *prompted* a general model. From roughly 2022 to 2024, you started to *chat* with instruction-following assistants, ask them to reason step by step, and feed them retrieved documents. And from about 2023 onward, you began to *wrap* models in harnesses that hand them tools, memory, code execution, browsers, shells, and explicit specifications.[3]

If you came in with a list of buzzwords like "prompt engineering" and "RAG," those are real and we will cover them. But a few themes matter just as much and rarely make the beginner lists: alignment and instruction following, tool use and function calling, structured outputs, prompt injection and agent security, evaluation benchmarks, multimodality, long-context engineering, and the protocols that wire tools together. Several of those turned out to matter more in practice than the buzzwords did.[4]

The story of AI, told in seven turns

The modern story starts with the **Transformer**, the neural-network architecture introduced in the 2017 paper *Attention Is All You Need*. It replaced the slow, sequential machinery of earlier models with "attention," a mechanism that lets a model weigh every word against every other word in parallel. That made models far easier to scale. The paper looked narrow and technical at the time, but it laid the foundation that general-purpose language models, then chat interfaces, then agents would all be built on.[5][6]

The second turn was learning *in context*. The 2020 paper behind GPT-3, *Language Models are Few-Shot Learners*, showed that one big model could perform many tasks straight from natural-language instructions and examples, usually with no retraining at all. That was the birth of prompting as a way to build software. When OpenAI shipped its API (application programming interface, the connection that lets one program call another) later that year, the

research finding became a product: text goes in, text comes out, and that is your programmable interface.[7]

The third turn was alignment, the work of making a raw model behave like a helpful assistant. The *InstructGPT* research showed that scale alone does not guarantee helpfulness, and that **reinforcement learning from human feedback (RLHF)**, training a model on human preference judgments, could make a smaller, better-behaved model more useful than a much larger raw one. Anthropic's *Constitutional AI* pushed further, steering behavior with an explicit set of written principles. This lineage is why we now have system messages, developer messages, policies, and published "model specs" that describe how an assistant should act.[9]

The fourth turn was reasoning through prompting. *Chain-of-thought (CoT) prompting* and *Self-Consistency* revealed something genuinely surprising: certain abilities were not just baked into the weights, they could be unlocked by *how* you prompted and decoded. Ask a model to think step by step, or sample several reasoning paths and take the majority answer, and accuracy jumps. Usage patterns turned out to be capability discoveries, and the first mass wave of "prompt engineering" was born.[11][13]

The fifth turn was grounding the model in knowledge it was never trained on. **Retrieval-augmented generation (RAG)**, introduced in a 2020 paper by Lewis and colleagues, gave a clean recipe: pair the model's internal memory with an external search step, so it can pull in fresh or private documents at answer time. This solved freshness and provenance at once, and it was usually cheaper and safer than retraining. RAG quickly grew from a research idea into an entire product category of vector databases, indexing pipelines, and "chat with your documents" tools.[14]

The sixth turn was the reason-and-act agent. *ReAct* made the core loop explicit: reason, take an action or call a tool, observe the result, repeat. *Toolformer* asked the next question, when should a model decide to call a tool at all? Then *Generative Agents*, *Reflexion*, and *MemGPT* pushed into persistent memory, self-reflection, and behavior that holds together over long stretches. By late 2024, Anthropic's widely read essay *Building Effective Agents* delivered the industrial lesson learned the hard way: the best real systems are usually simple, composable workflows with restrained agents, not grand autonomous beings.[16]

The seventh turn, the one you are living through, is agentic software engineering. Once models got good enough at code, and once terminals, editors, and file tools were exposed to them safely enough, "coding agents" became a whole new harness layer. Today's command-line tools, **CLI** meaning **command-line interface**, the text-based terminal where you type commands, are the clearest example. Claude Code, OpenAI's Codex CLI, Google's Gemini CLI, and the open-source Aider are not new *models*. They are *wrappers around models* that supply repository context, tool schemas, an execution environment, and review loops. That is why the

sharpest question in 2026 is often no longer "which model?" but "which model, inside which harness?"[20]

The papers that changed how people actually use models

This catalog leans toward *usage* breakthroughs, the results that changed what you do at the keyboard, not only the raw architecture underneath. For each one, there is a friendlier on-ramp than the paper itself.

Paper	Why it changed how you use AI	Friendlier on-ramp
Language Models are Few-Shot Learners (Brown et al., 2020)[7]	Made prompting and in-context learning a real alternative to task-specific training. "Write the instructions in plain text" became a serious idea.	OpenAI's original API launch was the bridge from research to building.[8]
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis et al., 2020)[14]	Introduced the "model plus retriever plus external corpus" pattern that became enterprise RAG. NLP here is **natural language processing**, the field of getting computers to handle human language.	Pinecone's RAG explainers are among the clearest lay introductions.[15]
Training Language Models to Follow Instructions with Human Feedback (Ouyang et al., 2022)[9]	Turned raw models into instruction-followers. This is the ancestor of every chat assistant, and the reason system and user messages matter.	OpenAI's prompt-engineering guide is the practical descendant.[22]
Chain-of-Thought Prompting Elicits Reasoning in Large Language Models (Wei et al., 2022)[11]	Showed that prompt style can unlock hidden reasoning ability, and kicked off the first big wave of reasoning prompts.	The DAIR.AI Prompting Guide is a strong, free starting point.[23]
Self-Consistency Improves Chain of Thought Reasoning (Wang et al., 2022)[12]	Showed that your *decoding* strategy is part of system design: sample several reasoning paths, then take the consensus. An early hint that spending more compute at answer time pays off.	Vendor prompting docs and the DAIR.AI guide popularized it.[23]
Constitutional AI: Harmlessness from AI Feedback (Bai et al., 2022)[10]	Mattered not because users read constitutions, but because it moved behavior control into explicit, written principles. This shaped public model specs.	OpenAI's Model Spec and Anthropic's published system prompts are the visible descendants.[24]

Paper	Why it changed how you use AI	Friendlier on-ramp
ReAct: Synergizing Reasoning and Acting (Yao et al., 2022)[16]	The canonical modern-agent paper. Interleaving thinking with actions became the default shape of every tool-using loop.	Lilian Weng's agent essay and Simon Willison's write-ups made it legible to builders.[25]
Toolformer: Language Models Can Teach Themselves to Use Tools (Schick et al., 2023)[17]	Made tool use a first-class skill rather than a bolt-on, treating calculators, search, and APIs as native parts of the workflow.	OpenAI's function-calling launch and Anthropic's tool-use docs were the industry versions.[26]
Generative Agents (Park et al., 2023)[18]	Seminal for long-term memory, reflection, and believable multi-agent behavior. It shaped later memory designs.	Lilian Weng's essay carried its planning-memory-tool triad to a wide audience.[25]
SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering (Yang et al., 2024)[20]	The clearest research bridge to CLI coding agents. It formalized the idea that models need a real interface to repositories, tests, and shells to fix real software. SWE is **software engineering**.	Simon Willison's writing on how coding agents work, plus the Codex CLI and Gemini CLI docs.[21]

For honorable mentions, add *Reflexion* for self-critique and episodic memory, *MemGPT* for memory that lives outside the context window, and the early *prompt injection* papers for the security lesson that agent builders learned the painful way.[19]

The architecture you can actually feel

You do not need to read a single architecture paper to use AI well. But it helps to know which under-the-hood changes you are *feeling* when a model gets faster, cheaper, or able to swallow your whole codebase. The simplest test for any architecture improvement is: did a user notice?

Improvement	What it changed for you	Representative work
Transformer self-attention	Made very large, general models possible at all. Without it, none of the rest exists.	The Transformer paper.[5]
Scale plus in-context learning	Let one model do many tasks from plain instructions. Prompting became a programming interface.	GPT-3 and the OpenAI API.[7]

Improvement	What it changed for you	Representative work
Instruction tuning, RLHF, constitutional tuning	Made models follow intent, refuse harm more consistently, and act like assistants instead of autocomplete.	InstructGPT and Constitutional AI.[9]
Diffusion and latent diffusion	On the image side, made "prompting an AI" mainstream before most people touched a text API. Latent diffusion cut the cost enough for everyone. DDPM is **denoising diffusion probabilistic models**.	DDPM, Latent Diffusion, DALL-E 2.[27]
RoPE and long-context methods	Let models handle very long prompts, which fed the "just give it the whole repo or transcript" workflow. RoPE is **rotary position embedding**, a way of telling the model where each word sits.	RoFormer, Position Interpolation, YaRN.[28]
KV cache	Stopped the model from recomputing everything on every word, making chat feel interactive. KV is **key-value**, the model's working memory for the current conversation.	Standard transformer decoding. [29]
MQA and GQA	Shrank that working memory so models answer faster and handle long context more cheaply. MQA is **multi-query attention**; GQA is **grouped-query attention**. You feel these as lower latency.	Fast Transformer Decoding, GQA, Mistral 7B.[30]
FlashAttention	Big speed and memory wins, especially on long sequences. Helped turn "long context" from a demo into a feature.	The FlashAttention papers.[31]
Mixture-of-experts (MoE)	Let total model size grow without paying full compute on every word, improving the quality-per-dollar curve. You feel it as stronger yet cheaper models.	Switch Transformers, Mixtral, DeepSeek V2/V3.[32]

Improvement	What it changed for you	Representative work
BF16 and FP8	Lower-precision math slashed training cost while keeping quality, enabling faster releases. BF16 is **bfloat16**; FP8 is **8-bit floating point**.	Mixed-precision and FP8 training work.[33]
Quantization	Made local inference and open-weight tinkering possible on modest hardware. This is why hobbyists can run capable models at home. GGUF is the **GPT-Generated Unified Format**, the popular local-model file format.	LLM.int8, GPTQ, QLoRA, the GGUF/llama.cpp ecosystem.[34]
LoRA and PEFT	Made customizing a model cheap, so you adapt instead of retraining from scratch. PEFT is **parameter-efficient fine-tuning**; LoRA is **low-rank adaptation**.	LoRA, QLoRA.[35]
Speculative decoding	Drafts several words with a small model and verifies with a strong one, cutting latency. You feel it as "faster model."	DeepMind's speculative sampling.[36]
Selective state-space models (Mamba)	An alternative to pure attention for very long sequences. Not yet the default, but important in the search for what comes after the Transformer.	Mamba.[37]
KV-cache compression	Directly attacked the cost of long context by shrinking that working memory, giving you cheaper, longer-running agents.	DeepSeek V2/V3 latent attention, TurboQuant.[38]
Structured outputs	Turned fragile free-text into dependable machine output, a quiet enabler of serious agents.	OpenAI Structured Outputs, Anthropic strict tool use.[39]
Prompt and context caching	Cut cost and latency for repeated long instructions, which matters enormously in production agents.	OpenAI prompt caching, Google context caching.[40]

Improvement	What it changed for you	Representative work
Native multimodality	Turned AI from a text box into something that can see, hear, and act, essential for browsing, computer use, and coding from screenshots.	GPT-4o, Gemini 2.0, computer use.[41]

The causal pattern underneath this table is worth holding onto: usage innovations arrive when the architecture creates enough slack. Context windows got longer, so people moved from pure prompting to RAG and then to "context engineering." Tool calls became reliable and schema-constrained, so people built agents instead of parsing chat text with regular expressions. Quantization and open weights matured, so local and CLI tools exploded. Once those usage patterns mattered, the labs started tuning models specifically for long context, tool use, code, and agent benchmarks. The loop feeds itself.[42]

The other timeline: when the architecture reached your hands

Everything above is the *idea* clock: papers and concepts in the order they were discovered. A second clock runs alongside it, the *product* clock, and for a user it matters more. A capability does not change your life when it is published in a paper. It changes your life when it ships inside a model you can actually call. So this section runs the model releases that moved the floor, and pairs each one with the architectural lever it first put in your hands and the new thing it let you do. You will never build any of these. But knowing which release introduced "thinking," or "see my screen," or "swallow the whole repository" is exactly what tells you which harness to reach for and how to drive it.

A warning before the table: this is a list of *inflection points*, not a changelog. Dozens of strong models shipped between these rows. The ones below are the releases where a behind-the-scenes architectural change first became something a normal person could feel.

Release (date)	The architecture it put in users' hands	What you could newly do
ChatGPT / GPT-3.5 (Nov 2022)	Reinforcement learning from human feedback, wrapped in a chat loop	Hold a real back-and-forth conversation. This is the release that made AI mainstream overnight.[43]
GPT-4 (Mar 2023)	A large capability jump, plus image input and reliable tool calling	Trust the model on harder reasoning, and start wiring it into software and tools instead of just chatting.[44]

Release (date)	The architecture it put in users' hands	What you could newly do
Claude 100K, then 200K context (May, Nov 2023)	Long-context engineering reaching production	Drop a whole codebase, contract, or research paper into a single session instead of chopping it into pieces.[45]
GPT-4o and Gemini 1.5 (2024)	Native multimodality and a one-million-token window	Talk to it, show it screenshots, and "just give it everything" without building a retrieval pipeline first.[46]
Claude 3.5 Sonnet (Jun 2024), then **Computer Use** (Oct 2024)	A mid-tier model beating the previous flagship, and the first model to drive a graphical interface	Stop paying for the biggest model by default, and let an agent actually click and type in a real screen.[47]
OpenAI o1 (Sep 2024)	Inference-time compute: a model trained to "think" before answering	Spend extra compute on hard problems on demand. Reasoning effort became a dial you control.[48]
DeepSeek V3 and R1 (Dec 2024, Jan 2025)	Open-weight mixture-of-experts at scale, and an open reasoning model, trained for a fraction of the usual cost	Run frontier-grade reasoning cheaply or locally. The "DeepSeek moment" collapsed the price floor and rattled the whole market.[49]
Claude 3.7 Sonnet and Claude Code (Feb 2025)	Hybrid reasoning (standard or extended thinking) in one model, plus a first-class CLI coding harness	Toggle "think harder" per task, and move serious coding out of the chat box and into the terminal.[50]
Claude 4 and the Opus 4.x line (May 2025 onward)	Models tuned specifically for long-horizon agentic work	Hand an agent a multi-step job and let it plan, run tests, and iterate for hours with light supervision.[51]
GPT-5 (Aug 2025)	A unified system that routes between a fast model and a thinking model automatically	Stop picking models by hand for routine work. The system decides how hard to think for you.[52]
Gemini 3 Pro (Nov 2025)	Trillion-scale sparse mixture-of-experts, a one-million-token window, and a built-in agentic coding environment	Reason across an entire data room in one shot. Its launch set off a frantic late-2025 release sprint across every lab.[53]

Release (date)	The architecture it put in users' hands	What you could newly do
The Chinese open-weight wave (2025 to Apr 2026)	Permissively licensed mixture-of-experts models using latent and sparse attention (DeepSeek V4, GLM-5.1, Kimi K2.6, MiniMax, Qwen 3.6)	Get near-frontier coding and reasoning at five to thirty times lower cost, and self-host it if you want to.[54]
The mid-2026 frontier (GPT-5.5, Claude Opus 4.8, Gemini 3.1)	Refinement over revolution: reliability, honesty about their own mistakes, and longer stretches of autonomy	Trust agents on longer unattended runs. The jumps are smaller now, but they land every few weeks instead of every year.[55]

Read down that table and the architecture section snaps into focus, because each lever shipped on a specific day. Long context stopped being a demo when Claude hit 100K and Gemini reached a million tokens, which is the moment "just paste the whole thing" beat careful retrieval for many tasks. Mixture-of-experts stopped being a research curiosity when Mixtral and then DeepSeek made strong-yet-cheap models real, which is why a hobbyist can now run a capable model at home. Inference-time compute became usable the day o1 and hybrid-reasoning Claude let you ask for more thinking, which is why "reasoning effort" is now a setting and not a paper. And open weights plus aggressive quantization, delivered by the DeepSeek and broader Chinese wave, are the direct reason local and CLI tooling exploded: once a frontier-adjacent model fit on your own hardware, the terminal became a serious place to work.[56]

Two practical lessons fall out of this clock. First, the cadence has gone vertical. Releases used to be yearly events; now they arrive every few weeks, and the version numbers have largely stopped meaning anything, so chasing the top of a leaderboard that reshuffles weekly is a losing game. Pick a model by whether it fits your task and your harness, not by who held the crown on Tuesday.[57][58] Second, *how* you reach a model is becoming its own decision. Aggregators and gateways such as OpenRouter, Vercel AI Gateway, and Helicone let you swap one provider for another with a single configuration change, and search-native products like Perplexity wrap models in a different default loop entirely. The provider is turning into a routing choice rather than a marriage, which is good news for you and uncomfortable news for anyone betting on lock-in.[59]

Ideas rise, and ideas fade

It is tempting to read the story so far as pure accumulation, one clever feature piled on the last. That is half the picture. The other half is that ideas *die*. Every advance above quietly retired something that came before it, and the techniques you are being taught right now

include a few that working builders are already easing out the door. This is the part beginners rarely get, because books and courses always lag the practice by a year. "Fading" almost never means "gone." It means "no longer the default, and a little embarrassing if you reach for it without thinking."

Here is the succession laid out plainly. Read it as a set of trades, not a tier list.

What is fading	What is taking its place	Why the swap happened
Fine-tuning a model for every task	Prompting one general model	GPT-3 showed you could describe the task in plain text instead of training for it.
Elaborate prompt tricks and bribery ("I'll tip you $200")	Plain, clear instructions	Models got good enough at following intent that the theatrics stopped earning their keep.
Treating the prompt as the whole job	Context engineering	What the model needs is the prompt *plus* the right documents, history, and tool output, assembled deliberately.
Naive chunk-and-stuff vector-database retrieval	Long context, agentic search, and contextual memory	Bigger windows, file search, and accumulating memory beat a brittle one-shot index for many jobs.
Bolt-on tool wrappers and regex-parsing chat text	Native function calling and structured outputs	Once tool calls were reliable and schema-constrained, you could build on them instead of around them.
Loading every tool definition into the prompt	Code execution and progressive tool discovery	Stuffing dozens of schemas into context was burning the window before work began.
Dense models where every parameter fires on every token	Mixture-of-experts	You get more total capacity without paying full compute on each token.
Single-shot answers	Reasoning models with a thinking dial	Spending compute at answer time turned out to buy real accuracy on hard problems.
Cloud-only frontier access	Open weights, local inference, and CLI harnesses	Capable open models that fit your own hardware moved serious work to the terminal.
Re-retrieving the same knowledge every query	A compiled knowledge base that accumulates	Compile once and keep it, instead of re-deriving your own context on every call.

Three of those trades are worth a closer look, because they are the ones changing how people work *this year*. None of them is "X is dead," and you should distrust anyone who says it that simply.

Retrieval is the loudest example. The 2023 recipe, chop documents into chunks and shove them into a vector database, is the version people now bury, and "RAG is dead" went viral in early 2026. It is clickbait. Retrieval as a market is still growing fast.[70] But naive chunk-and-stuff retrieval fails often enough in production that builders moved on, for three reasons: context windows got big enough to sometimes just hand the model everything, agents learned to search for what they need with plain text search ("grep") over files, and "contextual memory" that adapts across sessions started replacing one-shot lookups.[71] The most capable coding agents now lean on a small always-loaded index that points to topic files pulled on demand, backed by lexical search, rather than a mandatory vector database.[72] If you are learning retrieval today, learn hybrid search, reranking, and evaluation, not the toy pipeline.

Tool plumbing is the second. The **Model Context Protocol (MCP)**, Anthropic's late-2024 standard for plugging tools into a model (often called a USB-C port for AI),[76] succeeded fast. But every connected server's tool definitions load into the context window on every request, so a few busy servers can eat most of your window before the model reads your question.[73] The fix that took off in 2026 is "code mode": expose the tools as code files and let the model write a small program that calls only the ones it needs. Anthropic reported one workflow dropping from roughly 150,000 tokens to about 2,000, and the reasoning is almost funny, models trained on mountains of real code are simply better at writing code than at emitting a special tool-call format.[74] MCP is not dying. The naive all-schemas-in-context habit is.[75]

Knowledge storage is the third. In April 2026 Andrej Karpathy popularized a "compile, don't retrieve" pattern: instead of re-fetching raw documents on every query, have the model build and maintain a structured Markdown knowledge base from your sources, with a schema file telling it how to behave. The slogan that sold it: retrieval forgets, a wiki accumulates and compounds.[77] For one person or a small team this often beats a retrieval pipeline, and it fits how CLI coding agents already read project files.[78] That convergence is not a coincidence. The terminal, where a model can read files, run commands, and get real feedback, is where day-to-day AI work is migrating, and "agentic engineering," humans directing agents rather than typing every line, is now a named practice rather than a novelty.[79]

One trade runs the other way and costs you instead of saving you: the security bill. The moment you give an agent private data, expose it to untrusted text, and let it talk to the outside world, you have built what Simon Willison named the "lethal trifecta," and a single poisoned web page or email can turn a helpful assistant into a data-exfiltration tool. There is still no reliable fix. The working rule is to never let one unsupervised agent hold all three capabilities at once.[80] If you remember one safety idea from this guide, remember that one.

The practical playbook that falls out of all this is short. Give the model the *right* context, not the most context. Let an agent search and read files rather than pre-indexing everything. Keep a compiled, durable knowledge base instead of re-explaining yourself every session. Move serious work to a CLI harness where the model can act and get feedback. Pick models by fit, not by leaderboard. And assume any agent touching untrusted content can be hijacked, so design as if it will be.

Choosing your model: inside one lab, and across the field

The release timeline tells you that capabilities arrive on dates. It does not tell you which model to actually pick today, which is the question you hit the first time you face a dropdown with eight options. There is no single best model. There is only the best model for *this* task under *your* constraints, and working builders have mostly converged on the same short method for finding it.[60]

Start by throwing out anything that fails a non-negotiable, before you compare quality at all. Three constraints do most of the eliminating: privacy (can this data leave your machine or your region?), latency (does your use tolerate a slow thinker?), and cost at your actual volume. Whatever survives, fit to a capability tier. Every major provider now sells the same three-rung ladder under different names. A small, fast tier (Claude Haiku, GPT nano and mini, Gemini Flash-Lite) handles classification, extraction, formatting, and high-volume routine work. A mid tier (Claude Sonnet, GPT mid, Gemini Pro) is the daily driver for most coding, writing, and production traffic. A top tier (Claude Opus, GPT Pro and thinking, Gemini Pro and Ultra) earns its higher price only on the hardest reasoning, the long-horizon agent runs, and the gnarly code.[61] Layered on top of that ladder is the reasoning dial: the same model can often be told to think longer on a hard problem, which is a setting you choose per task rather than a separate product.

Two traps catch beginners here. The first is that the cheapest model per token is not the cheapest model per task. A weak model that needs three retries, produces sloppy output, or wastes context can cost more than a stronger one that nails it the first time, so measure cost per finished unit of work, not per million tokens.[62] The second is trusting public leaderboards as a verdict. They are a useful first filter, but once you have two or three finalists, the only test that matters is running them on your own examples.[60]

Now the part most guides skip, and the part you specifically want: the labs have different philosophies, and those philosophies leak into the models in ways you can feel. Reading them is a genuine selection tool. Anthropic grew out of a 2021 split from OpenAI over how seriously to take safety, and that lineage shows up as Constitutional AI (training a model to critique itself against written principles), heavy investment in interpretability, and a product line that leans toward careful refusals, honesty about its own mistakes, and strong coding and long-context work. It has earned a lot of enterprise and regulated-industry trust on exactly those

grounds.[63] OpenAI's bet is breadth and "it just works": the widest task coverage, the most mature tooling ecosystem, the deep Microsoft partnership, and a routing philosophy (GPT-5 picks an internal model for you) that spares you the choice. It even reversed its closed-only stance with open-weight gpt-oss models.[64] Google brings research depth and raw infrastructure scale, which surface as the strongest native multimodality, the largest context windows, and tight integration across Google's own products. The Chinese labs (DeepSeek, Alibaba's Qwen, Z.ai's GLM, Moonshot's Kimi, MiniMax) share a strategy of efficiency and open weights, doing more with less and undercutting on price, which is why they own the "download it and run it cheaply" niche.[63] xAI's Grok leans into speed, real-time data from X, and a less-filtered personality, on a fast and noisy release cadence. And a growing set of open-weight Western models (OpenAI's gpt-oss, Mistral, NVIDIA's Nemotron) exists for teams that want to self-host without going offshore.

So match the lab to your priority. Auditability and caution point one way, broadest ecosystem and hands-off routing another, maximum context and Google integration a third, and ownership, privacy, or raw cost a fourth. Then refuse to marry any of them. Gateways and aggregators let you route per request and swap providers with a one-line change, so hard-coding a single model name into your workflow is just technical debt that compounds every few weeks as the frontier reshuffles.[59]

Prompting that actually works

Prompting is the cheapest lever you have, and most people waste it by being vague. The fixes are dull and they work. Be specific about what you want and what you do not want. Show an example or two of a good answer, because demonstrating beats describing. Ask for step-by-step reasoning on anything with logic in it. State the format and the length, since "summarize this" and "summarize this in five bullet points for a chief financial officer" produce very different results. And use the role separation the tools hand you: a system or developer message sets the durable rules and persona, while the user message carries the specific request.[22]

The bigger move, the one that marks someone who has left beginner territory, is the shift from prompt to context. The elaborate tricks of 2023 (threats, fake tips, "you are a world-class expert," promising the model a reward) mostly stopped earning their keep as models got better at following plain intent. What replaced them is context engineering: deliberately assembling the whole information package the model needs, the instructions plus the right documents, the relevant history, and the tool outputs, instead of agonizing over the wording of one sentence.[81] Give the model the right context, not the most context. A prompt bloated with irrelevant material makes answers worse, not better, because the model has to dig the signal out of your noise.

Why harnesses and tools exist, and how to drive them

A bare language model is a brain in a jar. It is brilliant at producing text, but on its own it cannot look anything up, run a calculation it is unsure about, remember yesterday, read your files, or check whether its own answer was right, and it cannot reliably tell you when it is guessing. Almost every frustration you have felt (it invented a citation, it used stale facts, it lost the thread of a long task) traces straight back to the jar.[16]

Tools are how you open the jar. A tool is just a function the model is allowed to call: search the web, run code, query a database, read or write a file, hit an API. Give it a calculator and it stops fumbling arithmetic; give it search and it stops inventing facts; give it a code sandbox and it can test its own work instead of hoping. The advance that made this dependable was structured tool calling, where the model emits a clean, schema-constrained request instead of free text you have to parse, and that is the quiet foundation under every reliable agent.[26][39] The plumbing matured quickly: a shared protocol for connecting tools (MCP) crossed tens of millions of installs and was handed to a neutral foundation, which is the unglamorous sign of a real ecosystem.[65]

A harness is the scaffolding that wraps a model and its tools into something that does work over time. It supplies the loop (reason, act, observe the result, repeat), the memory, the file access, the execution environment, and the guardrails. This is why the sharp question is rarely "which model?" but "which model, in which harness?" A CLI coding tool like Claude Code or Codex CLI is a harness: it hands the model your repository, your shell, and your tests, then runs the loop against real feedback until the work passes. Driving one well is mostly supervision and scope. Give it a clear, bounded task, let it act, watch the diffs, and keep it away from anything irreversible without your sign-off.[21] The judgment on adding a tool is the same as the judgment on adding context: include the few that genuinely help, not every one you can, because each tool you expose is one more thing for the model to reason about and one more thing that can go wrong.

Orchestration: when models direct other models

The newest layer, and the one most in flux, is orchestration: using a model to coordinate other models. Once a single agent could plan and act, the obvious next step was to let it delegate, and that is where the field is experimenting hardest right now.

The headline pattern is orchestrator-worker, also called orchestrator-subagent or hub-and-spoke. A lead agent takes the goal, breaks it into independent pieces, spawns several specialized subagents to work those pieces in parallel (each with its own fresh context window, which is the crucial trick), then verifies and stitches their results into a single answer, usually with a separate checking pass.[66] It maps cleanly onto how a good manager works: plan, delegate to specialists, check the work, integrate. Anthropic's own research feature is

built this way and beat a single strong agent by about 90 percent on its internal evaluation, at a cost of roughly fifteen times the tokens.[66]

That cost ratio is the whole lesson in one number. Agent architecture is a token-spending strategy, and the orchestrator-worker topology is expensive, justified only when a task genuinely splits into independent, breadth-first parts such as researching across many sources or reviewing a large codebase from several angles. For most work, one well-driven agent is cheaper and better.[67] The fuller toolkit, drawn from Anthropic's widely used taxonomy, is six patterns you mix and match: prompt chaining (sequential steps with checks between them), routing (classify the request, then send it to the right handler), parallelization (run independent subtasks at once), orchestrator-workers (delegate and synthesize), evaluator-optimizer (one model drafts, another critiques, the first revises, in a loop), and the fully autonomous agent for open-ended tasks.[68] That evaluator-optimizer loop, where one model prompts and grades another model's output and feeds the critique back, is the iterative pattern most people mean when they say "have the AI check its own work."

If you take nothing else from this section, take the three reliability habits that pay off even without a multi-agent system: externalize state to a file or memory before your context window fills, give each worker a self-contained task description so it does not lean on hidden context, and verify high-stakes outputs (citations, code, factual claims) with a separate pass. Most of the reliability win of orchestration comes from those three, at none of the cost.[67] And keep the caution in view. Multi-agent setups are powerful, expensive, and still experimental, so start with one agent and reach for a fleet only when the task truly demands it.[69]

From buzzword to literature: a concept map

Some of the terms you will hear have a single canonical paper behind them. Others are engineering doctrines that got named *after* the fact, where the honest move is to point at the closest research anchor *and* the practical gateway, rather than pretend the literature has settled.

Theme	Canonical paper or closest anchor	Practical gateway
Prompt engineering	No single canonical paper; closest anchors are GPT-3 and the CoT work.[7]	OpenAI's prompting guide; Anthropic's prompting docs; the DAIR.AI guide.[22]
System and user prompts	Closest anchors: InstructGPT and Constitutional AI.[9]	OpenAI's Model Spec; Anthropic's published system prompts.[24]
RAG	Canonical: Lewis et al., 2020.[14]	Pinecone's RAG series.[15]

Theme	Canonical paper or closest anchor	Practical gateway
Context engineering	No single paper; lineage is RAG plus long memory plus agent loops. Named by Anthropic and Karpathy in 2025.[81]	Anthropic's "Effective context engineering for AI agents."[81]
Tool use / function calling	Canonical: Toolformer. Practical anchor: ReAct.[17]	OpenAI function calling; Anthropic tool-use docs.[26]
Agentic harnesses	Canonical: ReAct. Industrial distillation: Anthropic's agent taxonomy.[16]	Lilian Weng's essay; Anthropic's "Building Effective Agents."[25]
Autonomous agents	Canonical: ReAct, Generative Agents, Reflexion.[18]	Lilian Weng's essay is the clearest bridge.[25]
CLI coding tools	Closest anchor: SWE-agent and SWE-bench.[20]	Simon Willison on coding agents; Codex CLI, Gemini CLI, Aider docs.[21]
Wiki-based memory	No single paper; closest anchors are Generative Agents, MemoryBank, MemGPT.[82]	Karpathy's LLM-wiki gist and the follow-on guides.[77]
Spec-driven development	No canonical paper yet; mostly an engineering doctrine. Closest neighbor: SWE-agent.[20]	GitHub's Spec Kit and Martin Fowler's analysis.[83]
Prompt injection and agent security	Canonical: indirect prompt injection and attacks on LLM-integrated apps.[84]	Simon Willison's essays did the public sense-making.[80]
MCP and tool ecosystems	No foundational paper in the RAG/ReAct sense; this is a standards-layer development.[85]	Anthropic's MCP announcement and docs.[85]
Evaluation benchmarks	Anchors: SWE-bench, GAIA, WebArena, MT-Bench. GAIA is **General AI Assistants**. These changed what labs optimize for.[86]	Benchmarks spread via papers first, then model cards and vendor docs.[86]

If you want the shortest possible thesis for this whole map: the field moved from *better models* to *better interfaces for using models* to *better systems around models*.[87]

A reading path for the impatient

The best order is not oldest to newest. It is scaffolding first, operational depth second.

Start with the usage shift itself. Read GPT-3 and then OpenAI's original API framing, because together they give you the mental jump from "train a model for a task" to "describe the task in text."[7] Next read InstructGPT and Constitutional AI, which explain why a chat assistant differs from a raw model, and why behavior is partly a product of tuning and policy.[9] Then read Chain-of-Thought and Self-Consistency, where you learn that inference strategy and prompt structure are themselves part of capability.[11]

Now read RAG, the cleanest bridge from pure prompting to production systems.[14] Follow it with ReAct and Toolformer, the doorway to agents and tool use; once those click, most agent frameworks read as variations on a known theme.[16] Then the memory cluster, Generative Agents, Reflexion, and MemGPT, which is how researchers think about planning, episodic memory, and long-horizon continuity.[18]

Finally, for software work specifically, read SWE-agent and SWE-bench, then pair them with the current practitioner writing on coding agents and the karpathy-wiki pattern. That is where the frontier of everyday AI use actually sits in 2026.[20]

The one-sentence version of the entire story: Transformers made giant general models possible; GPT-3 made them usable through prompts; instruction tuning made them conversational; CoT and RAG made them more capable and grounded; ReAct and tool use made them act; memory and context engineering made them persist; and coding harnesses made them collaborators instead of merely eloquent text generators.[88]

Open questions and honest limits

A few topics simply do not have a single settled paper yet, and you should be suspicious of anyone who claims otherwise. System prompts, context engineering, wiki-based memory, MCP, and spec-driven development are shaped at least as much by platform docs, engineering blogs, and open-source practice as by formal research. For those, "closest academic anchor plus practical gateway" is the truthful framing.[89]

A couple of terms live right on the boundary between research and tooling. The "IQ quant" formats you will see in local-model circles are tied to the llama.cpp and GGUF ecosystem rather than one classic paper. And TurboQuant, which keeps coming up in long-context discussions, is a genuine and recent research result about compressing the model's working memory, not yet an industry-standard fixture on the level of established methods like GPTQ or QLoRA.[90]

The biggest unresolved question in the field is also the most practical one for you: will future progress come mostly from better base models, or from better scaffolding around them? The most defensible answer today is "both, in a tight feedback loop." The labs are clearly still improving models, but real-world capability increasingly depends on retrieval, tool design, memory, structured outputs, evaluation, and harness quality. Which is the entire reason this guide spends as much time on harnesses as on the models themselves.[91]

A small glossary, for when the acronyms blur

AI, artificial intelligence. LLM, large language model. NLP, natural language processing. API, application programming interface. SDK, software development kit. RAG, retrieval-augmented generation. MCP, Model Context Protocol. CLI, command-line interface. RLHF, reinforcement learning from human feedback. CoT, chain-of-thought. MoE, mixture-of-experts. KV cache, key-value cache (the model's working memory for the current conversation). MQA and GQA, multi-query and grouped-query attention. RoPE, rotary position embedding. LoRA, low-rank adaptation. PEFT, parameter-efficient fine-tuning. QLoRA, quantized LoRA. GGUF, GPT-Generated Unified Format (the common local-model file format). BF16 and FP8, sixteen-bit "brain float" and eight-bit floating-point number formats. DDPM, denoising diffusion probabilistic models. SWE, software engineering. GAIA, General AI Assistants (a benchmark). And the "lethal trifecta," the dangerous combination of private data, untrusted input, and an outbound channel in a single agent.[93]

Notes and references

1. This three-layer framing (models, interfaces, harnesses) is a synthesis rather than a single cited claim, but it tracks the actual sequence of papers and product releases laid out in the rest of these notes. *back*
2. The progression from pattern recognition to planning agents is the connective tissue of the citations below; no single source owns it. *back*
3. For the "wrap the model in a harness" end of this timeline, the clearest practitioner statement is Anthropic, *Building Effective Agents* (Dec 2024): anthropic.com/engineering/building-effective-agents. *back*
4. The security and evaluation themes in particular are easy to skip as a beginner and expensive to skip as a builder. See the prompt-injection and benchmark notes below. *back*
5. Vaswani et al., *Attention Is All You Need* (2017). arXiv: 1706.03762. *back back*
6. The title *Attention Is All You Need* went on to spawn roughly ten thousand imitators of the form "X Is All You Need," which is arguably a bigger contribution to academic culture than to academic rigor. Attention, it turned out, was not all anyone needed. They also needed data, money, and an electricity bill that would make a small nation flinch. *back*
7. Brown et al., *Language Models are Few-Shot Learners* (2020). arXiv: 2005.14165. *back back back back back*

8. OpenAI's original API announcement framed "text in, text out" as a programmable interface: openai.com/index/openai-api. *back*

9. Ouyang et al., *Training Language Models to Follow Instructions with Human Feedback* (2022). arXiv: 2203.02155. *back back back back back*

10. Bai et al., *Constitutional AI: Harmlessness from AI Feedback* (2022). arXiv: 2212.08073. *back*

11. Wei et al., *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models* (2022). arXiv: 2201.11903. *back back back*

12. Wang et al., *Self-Consistency Improves Chain of Thought Reasoning in Language Models* (2022). arXiv: 2203.11171. *back*

13. The discovery that adding "let's think step by step" to a prompt measurably raises accuracy means that, for a brief and glorious window, the cutting edge of computer science was politely asking the machine to try harder. Somewhere, a generation of strict computer-science professors felt a disturbance in the Force. *back*

14. Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks* (2020). arXiv: 2005.11401. *back back back back*

15. Pinecone's retrieval-augmented-generation learning series is a clear lay introduction: pinecone.io/learn/retrieval-augmented-generation. *back back*

16. Yao et al., *ReAct: Synergizing Reasoning and Acting in Language Models* (2022). arXiv: 2210.03629. For the industrial distillation, Anthropic, *Building Effective Agents*: anthropic.com/engineering/building-effective-agents. *back back back back back*

17. Schick et al., *Toolformer: Language Models Can Teach Themselves to Use Tools* (2023). arXiv: 2302.04761. *back back*

18. Park et al., *Generative Agents: Interactive Simulacra of Human Behavior* (2023). arXiv: 2304.03442. *back back back*

19. Shinn et al., *Reflexion: Language Agents with Verbal Reinforcement Learning* (2023). arXiv: 2303.11366. Packer et al., *MemGPT: Towards LLMs as Operating Systems* (2023). arXiv: 2310.08560. *back*

20. Yang et al., *SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering* (2024). arXiv: 2405.15793. Paired benchmark: Jimenez et al., *SWE-bench* (2023). arXiv: 2310.06770. *back back back back back*

21. Simon Willison writes prolifically and clearly on how coding agents work: simonwillison.net. Tool docs: Codex CLI (github.com/openai/codex), Gemini CLI (github.com/google-gemini/gemini-cli), Aider (aider.chat). *back back back*

22. OpenAI prompt-engineering guide: platform.openai.com/docs/guides/prompt-engineering. Anthropic prompting docs: docs.claude.com/en/docs/build-with-claude/prompt-engineering/overview. *back back back*

23. The DAIR.AI Prompt Engineering Guide: promptingguide.ai. *back back*

24. OpenAI Model Spec: model-spec.openai.com. Anthropic publishes its production system prompts in its release notes: docs.claude.com/en/release-notes/system-prompts. *back back*

25. Lilian Weng, *LLM Powered Autonomous Agents* (2023): lilianweng.github.io/posts/2023-06-23-agent. *back back back back*

26. OpenAI function-calling launch (June 2023): openai.com/index/function-calling-and-other-api-updates. Anthropic tool-use docs: docs.claude.com/en/docs/agents-and-tools/tool-use/overview. *back back back*

27. Ho et al., *Denoising Diffusion Probabilistic Models* (2020). arXiv: 2006.11239. Rombach et al., *High-Resolution Image Synthesis with Latent Diffusion Models* (2021). arXiv: 2112.10752. Ramesh et al., *Hierarchical Text-Conditional Image Generation with CLIP Latents* (DALL-E 2, 2022). arXiv: 2204.06125. *back*

28. Su et al., *RoFormer: Enhanced Transformer with Rotary Position Embedding* (2021). arXiv: 2104.09864. Chen et al., *Extending Context Window via Position Interpolation* (2023). arXiv: 2306.15595. Peng et al., *YaRN* (2023). arXiv: 2309.00071. *back*

29. The key-value cache is standard transformer decoding practice; Hugging Face documents it well: huggingface.co/docs/transformers/en/kv_cache. *back*

30. Shazeer, *Fast Transformer Decoding: One Write-Head is All You Need* (MQA, 2019). arXiv: 1911.02150. Ainslie et al., *GQA* (2023). arXiv: 2305.13245. Jiang et al., *Mistral 7B* (2023). arXiv: 2310.06825. *back*

31. Dao et al., *FlashAttention* (2022). arXiv: 2205.14135. *back*

32. Fedus et al., *Switch Transformers* (2021). arXiv: 2101.03961. Jiang et al., *Mixtral of Experts* (2024). arXiv: 2401.04088. DeepSeek-AI, *DeepSeek-V3* (2024). arXiv: 2412.19437. *back*

33. Micikevicius et al., *Mixed Precision Training* (2017). arXiv: 1710.03740. Peng et al., *FP8-LM: Training FP8 Large Language Models* (2023). arXiv: 2310.18313. *back*

34. Dettmers et al., *LLM.int8()* (2022). arXiv: 2208.07339. Frantar et al., *GPTQ* (2022). arXiv: 2210.17323. Dettmers et al., *QLoRA* (2023). arXiv: 2305.14314. The local ecosystem lives at llama.cpp: github.com/ggml-org/llama.cpp. *back*

35. Hu et al., *LoRA: Low-Rank Adaptation of Large Language Models* (2021). arXiv: 2106.09685. *back*

36. Chen et al., *Accelerating Large Language Model Decoding with Speculative Sampling* (2023). arXiv: 2302.01318. *back*

37. Gu and Dao, *Mamba: Linear-Time Sequence Modeling with Selective State Spaces* (2023). arXiv: 2312.00752. *back*

38. DeepSeek-AI, *DeepSeek-V2* (multi-head latent attention, 2024). arXiv: 2405.04434. Zandieh et al., *TurboQuant: Online Vector Quantization with Near-Optimal Distortion Rate* (2025, ICLR 2026). arXiv: 2504.19874. *back*

39. OpenAI Structured Outputs: openai.com/index/introducing-structured-outputs-in-the-api. Anthropic tool use (with strict schemas): docs.claude.com/en/docs/agents-and-tools/tool-use/overview. *back back*

40. OpenAI prompt caching: platform.openai.com/docs/guides/prompt-caching. Google context caching (Gemini): ai.google.dev/gemini-api/docs/caching. *back*

41. GPT-4o announcement: openai.com/index/hello-gpt-4o. Anthropic computer use: anthropic.com/news/3-5-models-and-computer-use. *back*

42. The "architecture creates slack, then usage rushes in to fill it" pattern is an editorial observation, but the citations above for long context, structured outputs, and quantization are its evidence. *back*

43. OpenAI launched ChatGPT, built on a GPT-3.5 model tuned with reinforcement learning from human feedback, on November 30, 2022. It reached an estimated 100 million users within two months, the fastest consumer-software ramp on record at the time. *back*

44. GPT-4 was released March 14, 2023, as a large multimodal model accepting text and image input. See OpenAI's overview: openai.com/index/gpt-4-research. Reliable function calling followed in June 2023 (see the function-calling note above). *back*

45. Anthropic was the first frontier lab to ship a 100K-token context window (Claude, May 2023), then 200K with Claude 2.1 (November 2023), enough to load entire codebases or contracts in one session. Timeline: scriptbyai.com/anthropic-claude-timeline. *back*

46. GPT-4o (May 2024) unified text, vision, and audio in one model: openai.com/index/hello-gpt-4o. Google's Gemini 1.5 brought a one-million-token context window and strong native multimodality the same year: blog.google/technology/ai/google-gemini-next-generation-model-february-2024. *back*

47. Claude 3.5 Sonnet (June 2024) was a mid-tier model that outperformed the prior flagship at lower cost, which reset how developers chose models. The October 2024 update introduced Computer Use, the first widely available model able to operate a graphical interface: anthropic.com/news/3-5-models-and-computer-use. *back*

48. OpenAI o1 (September 2024) was trained to reason before answering, making inference-time ("test-time") compute a usable dial: openai.com/index/learning-to-reason-with-llms. *back*

49. DeepSeek-V3 (December 2024) was a large open-weight mixture-of-experts model; DeepSeek-R1 (January 2025) was an open reasoning model trained at a small fraction of typical cost. Together they were the "DeepSeek moment." R1 paper: arXiv 2501.12948. V3 paper: arXiv 2412.19437. *back*

50. Claude 3.7 Sonnet (February 24, 2025) was Anthropic's first hybrid-reasoning production model, with a standard and an extended-thinking mode, shipped alongside the Claude Code research preview: anthropic.com/news/claude-3-7-sonnet. *back*

51. The Claude 4 generation (Opus 4 and Sonnet 4, May 22, 2025) was tuned for long-horizon agentic work, and Claude Code reached general availability the same day: anthropic.com/news/claude-4. The Opus 4.x line (4.1, 4.5, 4.6, 4.7, 4.8) refined it through 2025 and 2026. *back*

52. GPT-5 (August 7, 2025) was framed as a unified system that routes automatically between a fast model and a deeper reasoning model: openai.com/index/introducing-gpt-5. *back*

53. Gemini 3 Pro (November 18, 2025) launched at the top of public leaderboards, built on a trillion-scale sparse mixture-of-experts with a one-million-token window and a built-in agentic coding environment. Its arrival reportedly triggered a competitive scramble across the other labs: blog.google/products/gemini/gemini-3. *back*

54. Between roughly April 7 and April 24, 2026, four Chinese labs shipped open-weight coding models in quick succession (Z.ai's GLM-5.1, MiniMax's M2.7/M3, Moonshot's Kimi K2.6, DeepSeek's V4 Pro and Flash), joining Alibaba's Qwen 3.x line. They reach near-frontier coding and reasoning at roughly five to thirty times lower cost. Round-up: artificialanalysis.ai/articles/deepseek-is-back-among-the-leading-open-weights-models-with-v4-pro-and-v4-flash. Architecture trends (multi-head latent attention, sparse and linear attention, multi-token prediction): magazine.sebastianraschka.com/p/a-dream-of-spring-for-open-weight. *back*

55. As of mid-2026 the closed frontier is OpenAI's GPT-5.5 (April 2026), Anthropic's Claude Opus 4.8 (May 2026), and Google's Gemini 3.1 Pro (February 2026), with each new point release emphasizing reliability and longer autonomy more than raw capability jumps. GPT-5.5: en.wikipedia.org/wiki/GPT-5.5. Claude release history: hidekazu-konishi.com/entry/anthropic_claude_model_release_timeline.html. *back*

56. This paragraph is the explicit join between the architecture table and the release table: each lever has a date it reached users. The supporting citations are the per-release notes above plus the architecture references earlier in this guide. *back*

57. The release cadence compressed from yearly to roughly biweekly across 2025 and 2026, with version numbers increasingly decoupled from capability. The Stanford AI Index and independent trackers show the gap between the top model and the tenth-ranked model narrowing sharply year over year, which is part of why no single ranking stays authoritative for long. *back*

58. xAI's Grok deserves special mention as the field's most reliable producer of brief, triumphant leaderboard cameos. A Grok release will occasionally vault to the top of a ranking and then get passed, sometimes within the same 24 hours, by whatever Google or OpenAI shipped the next morning. Treat any single leaderboard as a photograph of a horse race, not the final result. The horses are also on a treadmill that speeds up weekly. *back*

59. A structural shift worth noticing as a user: model aggregators and gateways (OpenRouter, Vercel AI Gateway, Helicone) now let you switch providers with a one-line configuration change, and search-native products like Perplexity wrap models in a different default loop. Perplexity's own 2026 move away from MCP internally is part of the same "the provider is a routing decision" story discussed in the tool-plumbing section. *back back*

60. The five-step method (eliminate on non-negotiables, classify by tier, read the right benchmarks, shortlist and run your own evaluations, then route rather than bet on one model) is laid out in iternal.ai's 2026 selection guide: iternal.ai/llm-selection-guide. The "context, cost, latency, compatibility" framing predicting fit better than parameter count is from a widely shared 2026 comparison: dev.to/superorange0707/choosing-an-llm-in-2026. *back back*

61. A practical task-to-tier decision tree (structured tasks to small models, reasoning to top tier, coding to the agentic-coding models, latency or cost to the cheapest that passes your bar) is maintained at: llmgateway.io/blog/how-to-choose-the-right-llm. For open-weight selection by license, GPU budget, and context, see: acecloud.ai/blog/best-open-source-llms. *back*

62. "The cheapest model per token is not the cheapest model per task" is the central finding of independent 2026 cost benchmarks that measure retry rates and output quality, not just sticker price: cloudzero.com/blog/llm-api-pricing-comparison, and Ian Paterson's "Brains per Buck" task-level testing: ianlpaterson.com/blog/llm-benchmark-2026-38-actual-tasks-15-models-for-2-29. *back*

63. On Anthropic's safety lineage and Constitutional AI, and OpenAI's breadth-and-ecosystem posture, see DataCamp's comparison: datacamp.com/blog/anthropic-vs-openai. On the three-way Western race and each lab's positioning: qverlabs.com/blog/openai-vs-google-deepmind-vs-anthropic-2026. On the Chinese labs' open-weight, efficiency-first national strategy: Rest of World, restofworld.org/2026/tiezhen-wang-china-us-open-source-ai, and the USCC report uscc.gov. *back back*

64. OpenAI broke from its closed-only history by releasing open-weight models (gpt-oss-120b and gpt-oss-20b) under the permissive Apache 2.0 license, as noted in the DataCamp comparison above: datacamp.com/blog/anthropic-vs-openai. *back*

65. The Model Context Protocol crossed roughly 97 million installs by March 2026 and, alongside OpenAI's AGENTS.md and Block's goose, anchors the Agentic AI Foundation formed under the Linux Foundation in December 2025, a sign that tool standards have moved from experiment to infrastructure: blog.mean.ceo/new-ai-model-releases-news-april-2026. *back*

66. Anthropic, *How We Built Our Multi-Agent Research System* (2025), describes the orchestrator-worker pattern (a lead agent plans and spawns parallel subagents, each with its own context, then synthesizes with a separate citation pass) and reports beating single-agent Claude Opus 4 by 90.2 percent on an internal evaluation: anthropic.com/engineering/multi-agent-research-system. *back back*

67. The roughly fifteenfold token cost of the multi-agent research setup, the "agent architecture is a token-spending strategy" framing, and the three reliability habits you can apply inside a single agent (externalize state, isolate workers, verify high-stakes outputs separately) are drawn from this analysis of the Anthropic system: theaiengineer.substack.com/p/how-anthropic-built-multi-agent-deep. On the gap between the blueprint and production reality (cost ceilings, accuracy targets, long-running state), see: fountaincity.tech/resources/blog/anthropic-multi-agent-blueprint-production. *back back*

68. The six agentic patterns (prompt chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer, and the autonomous agent) come from Anthropic's *Building Effective Agents*: anthropic.com/engineering/building-effective-agents. *back*

69. There is a well-documented rite of passage where your shiny new orchestrator, asked a one-line question, enthusiastically spawns fifty subagents, each of which spawns its own subagents, and your token bill briefly resembles a phone number. Anthropic's own team reportedly spent weeks teaching their agents not to do this. The lesson generalizes: an agent that can delegate will delegate, much like a middle manager who has just discovered they can forward emails. *back*

70. The "RAG is dead" headline is best read as half-true. For the clickbait-versus-reality framing, see the practitioner round-ups of early 2026, e.g. *RAG vs Long Context 2026* (byteiota): byteiota.com/rag-vs-long-context-2026-retrieval-debate, and *Should You Be Using RAG in 2026?* (dev.to), which notes the retrieval market still growing at roughly 49 percent annually: dev.to/riddhesh/should-you-be-using-rag-in-2026-28ef. *back*

71. For the "contextual memory surpasses naive RAG for agents" prediction, VentureBeat's 2026 data round-up: venturebeat.com/data/six-data-shifts-that-will-shape-enterprise-ai-in-2026. *back*

72. On long context plus lexical search ("grep") plus compaction displacing mandatory vector databases, including the layered always-loaded-index pattern used by capable coding agents, see AkitaOnRails, *Is RAG Dead? Long Context, Grep, and the End of the Mandatory Vector DB*: akitaonrails.com/en/2026/04/06/rag-is-dead-long-context. *back*

73. On MCP tool-discovery overhead eating the context window (with one reported deployment at 72 percent of the window consumed before the real prompt), see the 2026 MCP guide from Maxim: getmaxim.ai/articles/what-is-model-context-protocol-mcp-a-complete-guide-for-2026. *back*

74. Anthropic, *Code Execution with MCP: Building More Efficient Agents* (Nov 2025): anthropic.com/engineering/code-execution-with-mcp. Cloudflare, *Code Mode: give agents an entire API in 1,000 tokens*: blog.cloudflare.com/code-mode-mcp. *back*

75. For the counter-case that MCP is maturing rather than dying, and that the backlash comes mostly from solo developers optimizing personal workflows, see *Is MCP Dead in 2026?* (Tyk): tyk.io/learning-center/is-mcp-dead-in-2026-why-enterprises-still-need-mcp. *back*

76. MCP is routinely described as "USB-C for AI," which is a lovely metaphor right up until you remember that USB-C is itself a sprawling family of incompatible cables that all look identical and only some of which charge your laptop. The metaphor may be more accurate than its authors intended. *back*

77. Andrej Karpathy's original LLM-wiki idea file (gist), April 2026: gist.github.com/karpathy/442a6bf555914893e9891c11519de94f. *back back*

78. For a step-by-step build of the pattern (raw sources, an LLM-maintained wiki, a schema file, and the ingest/query/lint operations), see the Starmorph guide: blog.starmorph.com/blog/karpathy-llm-wiki-knowledge-base-guide. The "compile, don't retrieve" insight connects directly to the file-native context-engineering literature, e.g. arXiv: 2602.05447. *back*

79. Simon Willison popularized "agentic engineering" as a name for humans directing agents rather than writing every line; see his ongoing writing at simonwillison.net. *back*

80. Simon Willison, *The Lethal Trifecta for AI Agents: Private Data, Untrusted Content, and External Communication* (June 2025): simonwillison.net/2025/Jun/16/the-lethal-trifecta. Meta's complementary "Agents Rule of Two" treats the three capabilities as a budget: an unsupervised agent may safely hold at most two. *back back*

81. Anthropic, *Effective Context Engineering for AI Agents* (2025): anthropic.com/engineering/effective-context-engineering-for-ai-agents. Karpathy's June 2025 framing that "context engineering" beats "prompt engineering" as the core skill is discussed in the academic write-up at arXiv: 2604.04258. *back back back*

82. Zhong et al., *MemoryBank: Enhancing Large Language Models with Long-Term Memory* (2023). arXiv: 2305.10250. *back*

83. GitHub Spec Kit: github.com/github/spec-kit. Martin Fowler's site collects thoughtful analysis of LLM-assisted engineering: martinfowler.com. *back*

84. Greshake et al., *Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection* (2023). arXiv: 2302.12173. Liu et al., *Prompt Injection Attacks against LLM-integrated Applications* (2023). arXiv: 2306.05499. *back*

85. Anthropic's Model Context Protocol announcement (Nov 2024): anthropic.com/news/model-context-protocol. Spec and docs: modelcontextprotocol.io. *back back*

86. Jimenez et al., *SWE-bench* (2023): arXiv 2310.06770. Mialon et al., *GAIA: A Benchmark for General AI Assistants* (2023): arXiv 2311.12983. Zhou et al., *WebArena* (2023): arXiv 2307.13854. Zheng et al., *Judging LLM-as-a-Judge with MT-Bench* (2023): arXiv 2306.05685. *back back*

87. "Better models, then better interfaces, then better systems" is the editorial spine of this guide; the citations throughout are its support. *back*

88. This single-sentence history compresses the citations from the seven turns above. If a beginner memorizes only one sentence, this is the one to memorize. *back*

89. The honest-limits framing matters because beginners often assume every named technique has a definitive paper. Many of the most useful 2026 practices live in blogs, docs, and gists, which is exactly why this reference list mixes arXiv links with engineering write-ups. *back*

90. TurboQuant: arXiv 2504.19874. The "IQ quant" formats are a feature of the llama.cpp/GGUF tooling rather than a single paper: github.com/ggml-org/llama.cpp. *back*

91. The "both, in a tight loop" answer is supported by every harness-focused citation here, especially Anthropic's agent and context-engineering essays. *back*

92. Yes, the thing you learned last month is already slightly out of date. Welcome to the field. The good news is that everyone else is in the same boat, paddling furiously and pretending they read the latest paper. The other good news is that the *fundamentals* in this guide age much more slowly than the headlines do. *back*

93. If you made it all the way to the glossary, you are now legally entitled to nod knowingly when someone says "the lethal trifecta makes the context window untrustworthy," and to watch the rest of the meeting fall silent in respect. Results may vary. Side effects may include being asked to actually fix the security problem. *back*